

NRES 746, Week 6, Lecture 2: Find the Bugs

Name: _____

Instructions: Work alone for about 3 minutes, then compare with a neighbor. Keep laptops closed.

The code

Here are two short functions, possibly written with AI help. The first comes from today's live-coding problem (finishing Lab 4's toad monitoring design).

- `detect_prob(k)`: one person-day of searching detects an animal with probability `pobs`, so `k` person-days should detect it with probability $1 - (1 - p_{obs})^k$.
- `sim_detect(R)`: should simulate `R` surveys of a site where the frog from Lab 5 is present (3 visits per survey, detection probability 0.4 per visit) and return the fraction of surveys with at least one detection.

Each function below contains **exactly one bug**. Both run without an error, but give wrong answers.

```
detect_prob <- function(k, pobs = 0.02) {  
  1 - pobs^k  
}  
  
sim_detect <- function(R = 1000, visits = 3, p = 0.4) {  
  detected <- logical(R)  
  for (i in seq_len(R)) {  
    n_det <- rbinom(1, size = visits, prob = p)  
    detected <- n_det > 0  
  }  
  mean(detected)  
}
```

Part 1: Find and fix (individual, ~3 min)

For each function, circle the bug on the code above, write the corrected line, and say what would go wrong if you ran the buggy version.

1a. `detect_prob()`

1b. `sim_detect()`

Part 2: Catch it next time (with a neighbor, ~2 min)

2. For one of the bugs, what quick check (one line of code and the answer you'd expect) would have caught it?
